

Uncertainty Map

Colour is the temperature, fade is the confidence - and the page publishes the exact amount by which its own confidence is overstated.

PORT	PAGE FILE	BAND	COVERAGE
8513	app/phase2/uncertainty_page.py	plus-or-minus 2 sigma	91.2% against 95.4% nominal

A reconstruction that reports only a temperature invites a reader to trust every cell equally, and the cells are not equal. This page shows both numbers the model produces - what it thinks, and how much it should be believed - and then reports how well that second number has actually been checked.

1. What this feature is, in plain words

The model emits two things at every depth: a temperature, and a variance - its own estimate of how wrong that temperature might be. This page draws both at once. Colour carries the temperature; opacity carries the confidence, so an uncertain cell fades toward the background.

That double encoding has a flaw, and rather than ignore it we measured it and added a second view. **Fading a colour over a dark background pulls every hue toward that background**, so a low-confidence warm cell starts to read as a cool one - the fade corrupts the very temperature it is drawn on top of. The second mode drops the temperature entirely and shows uncertainty alone. It is also the more revealing picture: the bright, uncertain band sits over the Somali Current and the southern Arabian Sea eddy field, and the quiet Bay of Bengal interior is dark. **The model is least sure where the ocean is most active**, which is exactly what you would hope to see.

The raw model uncertainty was too confident, so it is corrected afterwards using a per-depth scaling fitted on floats from the **training** window and then reported on floats from the **test** window - two sets that do not overlap in time, so the report is not marking its own homework.

THE EVERYDAY PICTURE

A weather forecast that says '30 degrees' and one that says '30 degrees, give or take one' are different products. The second is more useful precisely because it admits its own limits - and this page goes further by telling you how often that 'give or take' has actually held.

2. What you actually see on the screen

Two view modes and one uncomfortable panel:

- **Temperature with confidence** - colour for the value, opacity for how much to lean on it.
- **Uncertainty alone** - the sigma field on its own scale, without hue contamination.
- The opacity range printed **in degrees Celsius** under the map, because sigma at 5 m and sigma at 1000 m are different sizes and a single fixed scale would make whole depths look uniformly vivid or uniformly faded.
- A calibration panel that reports coverage as a **range across depths**, never as a single flattering average.
- A provenance line stating whether the sigma being shown is calibrated - read from the record rather than assumed.

3. How to run it, step by step

RUN THIS

```
PYTHONPATH=src python -m streamlit run app/phase2/uncertainty_page.py --server.port 8513
```

- 1 Run the command and open **http://localhost:8513**.
- 2 Start in the combined view at 100 m. The audience sees temperature with confidence baked into it.
- 3 Switch to **uncertainty alone** and point at the Somali Current. Say the sentence: the model is least sure where the ocean is most active.
- 4 Read the printed opacity range in degrees Celsius aloud, and explain why it is per-view.
- 5 Open the calibration panel and read the coverage number - 91.2% against a nominal 95.4% - **out loud, before anyone asks**.
- 6 Move the depth to 100 m and note that both the model's largest uncertainty and its largest correction factor land there. Two independent signals, one depth.

RE-FIT AND RE-REPORT THE CALIBRATION

```
PYTHONPATH=src python scripts/phase2/calibrate_uncertainty.py
```

4. Why the problem statement needs this

The problem statement asks for a usable reconstruction, and a number without an error bar is not usable for a decision. A forecaster deciding on an evacuation needs to know whether the heat content figure in front of them is solid or shaky.

It is also the project's most direct statement of scientific character. A page about uncertainty that overstated its own uncertainty would be self-refuting, so this one publishes the gap between what it claims and what it delivers.

PS requirement	How this feature answers it
Usable, decision-ready output	Every temperature ships with a per-depth uncertainty a user can act on.
Honest evaluation	Coverage is measured on held-out floats and reported as a range across depths.
Independent validation	Calibration is fitted on train-window floats and reported on test-window floats - disjoint in time.

5. The measured numbers behind it

What is claimed, exactly, and what was measured [VERIFIED]:

Measurement	Value	How it is stated
Coverage of the plus-or-minus 2 sigma band	91.4% mean , range 79.7% to 96.2% by depth, on 909 held-out profiles	Reported as a range , never as the mean alone - the mean would hide an 80% depth at 50 m.
Nominal target	95.4%	So we are 4.2 points short . The band is called 'plus-or-minus 2 sigma' and is never labelled '95%'.
Where the model is least certain	100 m, median sigma 1.197 degC	Most certain at 500 m, 0.265 degC.
Largest correction factor applied	also at 100 m, x1.4583	Two independent signals pointing at the same depth: the raw model is least sure at the thermocline and was most overconfident there.
Calibration fitting set	3,423 train-window profiles	Coverage reported on 908 test-window profiles - disjoint in time.

- **The honest summary, in the project's own words: the uncertainty is improved, not calibrated.** The model remains mildly overconfident, and the freeze script checks that this claim wording has not been quietly strengthened - it is one of its eighteen checks.
- No plus-or-minus 1 sigma band and no confidence percentage appear anywhere in the interface, because neither would be defensible.
- The page's own acceptance test was found to be comparing two different quantities - dimensionless scale factors against a sigma in degrees Celsius - so it would have passed or failed for the wrong reason. It was replaced with a far stronger check that reconstructs the field twice, with and without calibration, and asserts depth by depth that the ratio equals the published factor, and that calibration moves sigma and **not** temperature.

6. The impact - and who it is for

Who	What changes for them
Forecasters	Knowing which parts of a field to act on and which to treat with caution - the difference between data and a decision.
Scientists	A basin map of where a satellite-driven reconstruction is structurally weak, which is itself a research finding.
The jury	A team publishing the exact size of its own overconfidence, before being asked.
Observation planners	One of the two inputs to Feature 16 - you send instruments where the model is uncertain.

WHY SOCIETY GAINS FROM THIS

Overconfident predictions cause worse decisions than honest uncertain ones. A public system that says clearly where it is weak lets a forecaster weight it correctly against other evidence, and lets a community understand why a warning carries the caveats it does.

7. Say this to the jury

The model produces two numbers at every depth - a temperature and how uncertain it is - and this page draws both. Switch to uncertainty alone and you can see the model is least sure exactly where the ocean is most active: the Somali Current and the Arabian Sea eddy field. Now the number I want to give you before you ask: our plus-or-minus 2 sigma band covers 91.2% of held-out observations against a nominal 95.4%. We are 4.2 points overconfident. We report coverage as a range by depth, 80.1 to 95.5, because the average would hide an 80% depth at 50 metres. And we never label the band '95%'.

Say the coverage gap unprompted. A page about uncertainty must not overstate its own.

8. Questions the jury will ask

Q. Your uncertainty is not properly calibrated, then?

A. Correct, and we use exactly that word. It is *improved*, not *calibrated*: post-hoc per-depth scaling brought us from clearly overconfident to mildly overconfident. The freeze check verifies that our claim wording has not drifted upward.

Q. Why not just report the average coverage of 91.2%?

A. Because it hides the worst depth. Coverage runs from 79.7% at 50 m up to 96.2%, and a user acting on a 50 m value deserves to know that band is the weakest one. Reporting a range costs us a nicer headline and gains a defensible one.

Q. How do you know the calibration is not fitted to the same data it is scored on?

A. The scaling is fitted on 3,423 profiles from the training window and reported on 908 profiles from the test window. The two sets are disjoint in time, and the calibration was fitted against the shipped checkpoint specifically - which the freeze script asserts.

9. The honest limit - say it before they find it

TWO THINGS THIS UNCERTAINTY IS NOT

It does not come from Monte Carlo dropout - the Phase-1 dropout uncertainty was measured overconfident by 1.6 to 3.5 times and is not used here. And it carries **no correlation between depths**: the model gives a variance per depth independently, so any quantity integrating across depths - such as the heat content in Feature 9 - inherits an uncertainty that is a measured lower bound rather than the true one. Both limits are stated on the pages that depend on them.

IN ONE LINE

Both of the model's numbers on one map - what it predicts and how much to believe it - with the exact size of its own overconfidence published.

91.2%

COVERAGE AGAINST A 95.4% NOMINAL

80.1 to 95.5%

THE RANGE, BECAUSE THE MEAN HIDES A BAD DEPTH

100 m

LEAST CERTAIN AND MOST OVERCONFIDENT DEPTH

Feature 13 of 17. Next: Feature 14.